

Pushing the Limits: Crafting an Immersive Mega-Canvas for Phish's Music Shows at Sphere™

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ACM Reference Format:

Guillaume Borgomano, Zane Kozak, Arnaud Grosjean, and Jörgen Steinheim. 2024. Pushing the Limits: Crafting an Immersive Mega-Canvas for Phish's Music Shows at Sphere™. In *Proceedings of SIGGRAPH Production Sessions '24*. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3641232.3649396>

1 Description

Moment Factory is excited to invite attendees to a production session at SIGGRAPH 2024, where we'll delve into the innovative production of "a famous band" concert series at a novel immersive venue. This venue, a multimedia megacanvas, is unlike any other, and our presentation will cover the collaborative effort, creative challenges, and technical breakthroughs achieved within a demanding three-month timeline.

Collaborative Efforts Behind the Scenes. Central to the success of this project was the tight collaboration between artists and technologists. This multidisciplinary cooperation was crucial in tackling the complexities of producing content for one of the largest and most sophisticated screen surfaces ever built. Our session will showcase the collective ingenuity and problem-solving that brought our vision to life.

Creative Challenges. In just three months, our team managed to conceptualize, design, produce, and integrate over 8 hours of custom content, working under the motto of "building the plane as we fly it." This required a rapid and flexible production approach. We'll discuss the creative and logistical hurdles of crafting four distinct shows that captured the band's spontaneous "jam" essence. These shows required a production style that could adapt dynamically to the band's improvisational performances, without relying on traditional time-coded cues. Content Strategy The venue's dome-like screen demanded content with exceptional detail in ultra-high resolutions, pushing our team to new creative heights. We developed a modular show-flow incorporating various thematic elements, from takeover



and graphic to abstract moments, designed to maximize pixel space usage. This strategy aimed to deliver a range of sensations and create immersive, illusionary effects. We'll outline how we balanced realtime and pre-rendered scenes and the pivotal role of Generative AI (Gen-AI) in accelerating our production pace and enhancing the visual narrative.

Technical Challenges. The project required pushing the limits of what's possible with cutting-edge technology. We'll explain how we utilized real-time engines like Notch and Unreal Engine to deliver smooth content at 10K resolution and 30fps. Our session will also cover the integration of Gen-AI-generated assets into our workflow, particularly through a multi-modular Diffusion model using ComfyUI. We faced numerous challenges with Gen-AI, including managing input types (images, texts, videos), structural control, upscaling techniques, and using detailers/refiners to enhance the detail in generated outputs.

Visual 'Jamming'. To align with the band's improvisational style, we developed a 'VJing-style' control toolbox. This toolbox, controllable externally through the grandMA lighting console, facilitated visual 'jamming' in sync with the band's live performances. We will discuss the effective use of this toolbox with layered pre-rendered content. Conclusion Our session aims to highlight Moment Factory's dedication to innovation in location-based and large-scale multimedia experiences. We hope to inspire fellow creators and technologists to explore new creative territories and push beyond the familiar.

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SIGGRAPH Production Sessions '24, July 27 - August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0514-4/24/07

<https://doi.org/10.1145/3641232.3649396>